MATH 243: Discussion Worksheet 9

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**Topics:** 15.1 Double Integrals over Rectangles; 15.2 Double Integrals over General Regions; 15.3 Double Integrals in Polar Coordinates

Please put away all electronic devices, including cell phones and calculators.



1. Evaluate each double integral.
   (a) $\iint_R \left(y + \frac{x}{y^2}\right) dA$, where $R = \{(x, y) \mid 0 \le x \le 2,\ 1 \le y \le 2\}$.
   (b) $\iint_R x\cos^2(y)\, dA$, where $R = [-2, 3] \times \left[0, \frac{\pi}{2}\right]$.
   (c) $\iint_R \frac{1}{(2x+3y)^2}\, dA$, where $R$ is the rectangle with vertices $(0, 1),\ (1, 1),\ (1, 2),\ (0, 2)$.

2. Find the volume of the solid that lies under the hyperbolic paraboloid
$$z = 3y^2 - x^2 + 2$$
and above the rectangle $R = [-1, 1] \times [1, 2]$.

3. Given the region $R$ bounded by the lines $y = 3x + 1$, $x = 1$, and $y = 1$, draw the region $R$ and set up the double integral $\iint_R f(x, y)\, dA$ as a Type I integral and as a Type II integral.

4. Evaluate $\iint_D y^2 e^{xy}\, dA$, where $D$ is the region bounded by $y = x$, $y = 4$, and $x = 0$.

5. Find the average value of the function $f(x, y) = xy$ over the triangular region with vertices $(0, 0)$, $(1, 0)$, and $(1, 3)$.

6. (a) Sketch the region of integration for $\int_0^1 \int_0^{y^2} f(x, y)\, dx\, dy$.
   (b) Reverse the order of integration for the integral in part (a).
   (c) Evaluate the integral from part (b) with the function $f(x, y) = \sqrt{x} + 3y$.

7. Using polar coordinates, find the volume of the solid that lies under the cone
$$z = \sqrt{x^2 + y^2}$$
and above the annulus $1 \le x^2 + y^2 \le 4$.

8. Evaluate $\int_0^{1/2} \int_{\sqrt{3}y}^{\sqrt{1-y^2}} xy^2\, dx\, dy$ by changing to polar coordinates.

9. Use polar coordinates to combine the sum
$$\int_{1/\sqrt{2}}^{1} \int_{\sqrt{1-x^2}}^{x} xy\, dy\, dx + \int_1^{\sqrt{2}} \int_0^x xy\, dy\, dx + \int_{\sqrt{2}}^{2} \int_0^{\sqrt{4-x^2}} xy\, dy\, dx$$
into one double integral, and evaluate the integral.

## SOME USEFUL DEFINITIONS, THEOREMS, AND NOTATION

**The Average Value of a Function of Two Variables.**
The average value of a function $f$ over a region $R$ of area $A(R)$ is

$$f_{\text{avg}} = \frac{1}{A(R)} \iint_R f(x,y)\,dA,$$

provided that $A(R) > 0$.

**Integrals over General Regions.** Suppose that $f$ is continuous on a region $D$.
If $D$ is a Type I region of the form

$$D = \{(x,y) \mid a \le x \le b,\ g_1(x) \le y \le g_2(x)\},$$

where $g_1$ and $g_2$ are continuous on $[a,b]$ and $g_1(x) \le g_2(x)$ for all $x \in [a,b]$, then

$$\iint_D f(x,y)\,dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x,y)\,dy\,dx.$$

If $D$ is a Type II region of the form

$$D = \{(x,y) \mid c \le y \le d,\ h_1(y) \le x \le h_2(y)\},$$

where $h_1$ and $h_2$ are continuous on $[c,d]$ and $h_1(y) \le h_2(y)$ for all $y \in [c,d]$, then

$$\iint_D f(x,y)\,dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x,y)\,dx\,dy.$$

**Double Integrals in Polar Coordinates.** Suppose that $f$ is continuous on a region $D$ in the plane, and that $D$ is described in polar coordinates by

$$\alpha \le \theta \le \beta, \qquad h_1(\theta) \le r \le h_2(\theta),$$

where $h_1$ and $h_2$ are continuous on $[\alpha,\beta]$, $0 \le h_1(\theta) \le h_2(\theta)$ for all $\theta \in [\alpha,\beta]$, and $0 \le \beta - \alpha \le 2\pi$. Then

$$\iint_D f(x,y)\,dA = \int_\alpha^\beta \int_{h_1(\theta)}^{h_2(\theta)} f(r\cos\theta, r\sin\theta)\,r\,dr\,d\theta.$$

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**Suggested Textbook Problems**
Section 15.1: 1-49, 53-56
Section 15.2: 1-50, 55-79
Section 15.3: 1-49